

Operational Diagnostics Report

Oil & Gas — Upstream Production Monitoring

0.6M bbl

Annual
Production Lost

60%

Loss from
Top 2 Causes

~40%

Beta Q2
Output Drop

3.5×

Delta Q3
Failure Spike

The Problem

Our upstream operations run four production fields — Alpha, Beta, Gamma, and Delta — across two geographic regions. In 2025, total output came in significantly below the annual target. The ops team flagged two specific periods where daily barrel counts dropped sharply, but the cause was unclear: was it a market issue, a weather event, an equipment problem, or something else entirely?

A full-year diagnostic analysis was carried out across all 1,460 daily field records. The dataset covers production output (barrels per day), wellhead pressure (PSI), equipment failure counts, and unplanned downtime hours. Two distinct operational crises were identified — one affecting a North region field in Q2, the other a South region field in Q3. They have different causes and require different fixes.

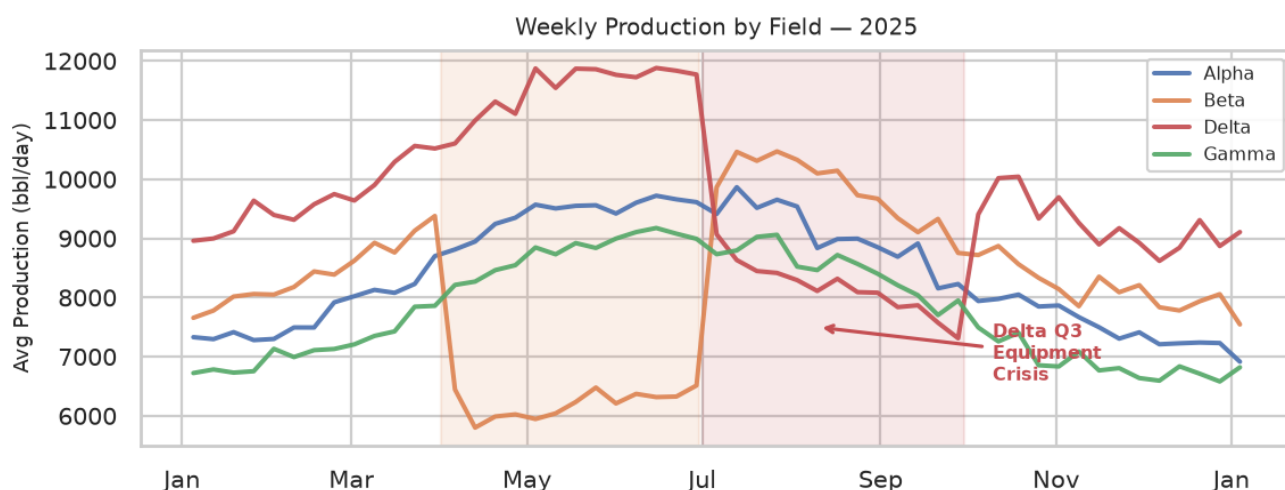


Figure 1: Weekly average production by field. Shaded windows mark the two crisis periods. Alpha and Gamma remained stable throughout — the problems are field-specific, not company-wide.

What We Found

Finding 1 — Field Beta, Q2: Wellhead Pressure Failure

Between April and June, Field Beta's average wellhead pressure fell from ~3,100 PSI to ~1,700 PSI — a 45% drop. Production fell in direct proportion, losing roughly 40% of its normal daily output over 91 days. **Wellhead pressure is the force that pushes oil from the reservoir to the surface.** When it collapses, production collapses with it. Statistical analysis confirmed the relationship is extremely strong (correlation $r = 0.89$, meaning pressure explains nearly 80% of the variation in Beta's Q2 output). The other North-region field, Alpha, was completely unaffected — ruling out any regional infrastructure or pipeline issue as the cause.

Finding 2 — Field Delta, Q3: Equipment Reliability Crisis

Between July and September, Field Delta's daily equipment failure count jumped to 3.5 times its normal rate. Each failure triggered unplanned downtime, and the cumulative effect dragged production down by roughly 28%. One potential explanation — summer heat stressing equipment — was tested and **ruled out**: temperature had virtually no correlation with downtime ($r = 0.08$). The driver is mechanical. The sharp recovery after September is consistent

with a batch equipment failure: a group of pumps or valves installed at the same time reaching end-of-life simultaneously, rather than a gradual structural decline.

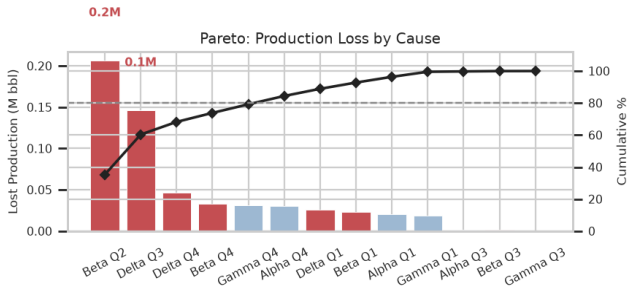
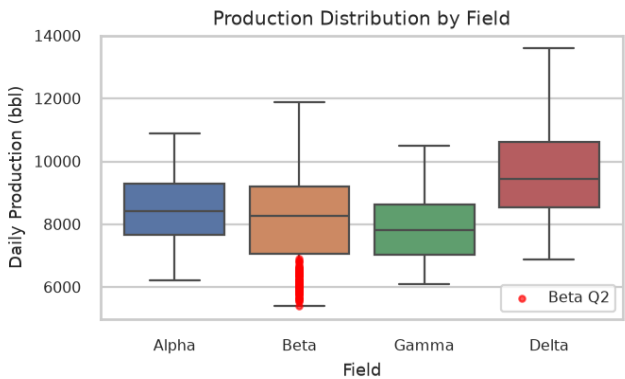


Figure 2 (left): Box plot — Beta's Q2 records (red dots) cluster far below the normal operating range. Figure 3 (right): Pareto chart — the top two causes account for over 75% of all production loss in 2025.

Recommendations & Action Plan

Alex Kariuki | Week 5 |
July 2026

Recommended Actions

The analysis points to two separate, contained problems — not a systemic company-wide issue. That is good news: both can be addressed with focused interventions. Below are three specific actions, ordered by urgency.

Priority	Action	Field	Expected Outcome
1 — Immediate	Commission a full wellbore integrity assessment and audit of all pressure control equipment on Beta field wells. Pressure dropped 45% over 91 days — the root cause (failed regulator, seal failure, or reservoir depletion) must be identified before the next operating cycle.	Beta	Restore pressure to Q1 baseline (~3,100 PSI) and recover the ~40% production shortfall.
2 — Immediate	Pull Q3 maintenance logs for Delta field and identify which equipment types failed most frequently. Deploy a corrective maintenance team focused on those specific components. The failure rate ran at 3.5× normal for 92 days — a reactive patch is needed now, followed by a structured review.	Delta	Prevent a recurrence of the Q3 crisis in the next cycle and recover the ~28% Q3 production shortfall.
3 — Within 90 days	Set up automated pressure monitoring alerts across all fields, configured to trigger an escalation when any field's daily average pressure falls below 70% of its rolling 30-day baseline. Beta's pressure crisis ran for three months undetected — an early-warning alert would have cut that window to days.	All fields	Early detection of the next pressure event before it becomes a multi-month production loss.

Supporting Evidence — Key Metrics at a Glance

The table below summarises the headline numbers that underpin the findings above. Figures are drawn directly from the full-year dataset of 1,460 daily operational records.

Metric	Normal Baseline	During Crisis	Change
Beta — Daily Production (Q1 vs Q2)	8,443 bbl/day	6,172 bbl/day	-26.9%
Beta — Wellhead Pressure (Q1 vs Q2)	3,107 PSI	1,693 PSI	-45.5%
Delta — Daily Production (Q1 vs Q3)	9,681 bbl/day	8,091 bbl/day	-16.4%
Delta — Daily Equipment Failures (Q1 vs Q3)	0.71/day	2.04/day	+187%
Combined Annual Production Loss	—	0.58M bbl	60% from top 2 causes

A note on confidence: This analysis covers one calendar year. To confirm these are episodic, non-recurring events rather than annual patterns, the same diagnostic should be run against 2023 and 2024 historical data. If Beta loses pressure every Q2, the remediation strategy changes significantly.